



**FRIEDRICH-SCHILLER-
UNIVERSITÄT
JENA**

INTERNSHIP PRESENTATION

FRIEDRICH SCHILLER UNIVERSITY JENA

FACULTY OF PHYSICS AND ASTRONOMY

INSTITUTE FOR SOLID STATE THEORY AND OPTICS

Effect of Localized Rotation Errors on the Deutsch–Jozsa Algorithm

Submitted in partial fulfillment of the requirements for the Internship Module

SUPERVISOR

Prof. Dr. Martin Gärttner

PROJECT ADVISOR

Niklas Euler

Presented by

Abiy Zelalem Tegegne

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Motivation of the Study

Why error location matters

Research problem

- Quantum algorithms depend on precise gates operations.
- A coherent over-rotation can accumulate systematically instead of averaging away.
- The same rotation may be harmless at one stage but destructive at another.

Study focus

- Compare localized $R_x(\theta)$, $R_y(\theta)$, and $R_z(\theta)$ errors.
- Test four physically meaningful locations, E_1 – E_4 .
- Identify the axis–location combinations that most reduce $P(0^n)$.

Motivation

Mechanism-matched mitigation requires knowing both what the error is and where it acts.

Objectives of the Study

The study separates axis, angle, and location

1 Model Implement coherent single-qubit rotations $R_x(\theta)$, $R_y(\theta)$, and $R_z(\theta)$.

2 Localize Insert each error separately at E_1 , E_2 , E_3 , and E_4 .

3 Measure Sweep θ and evaluate the probability $P(0^n)$.

4 Compare Identify robust and vulnerable stages of the algorithm.

The Deutsch–Jozsa Problem

$$f : \{0,1\}^n \rightarrow \{0,1\}$$

Boolean Function

Constant

$f(x)$ has the same value
for every input x .

Balanced

$f(x) = 0$ for half the inputs
and $f(x) = 1$ for the other half.

Task

Determine with certainty whether the function is constant or balanced.

Continued...

Quantum interference compresses many function values into one query

To decide with certainty

Classical deterministic(worst case)

$$2^{n-1} + 1$$

oracle queries in the worst case

Deutsch–Jozsa quantum algorithm

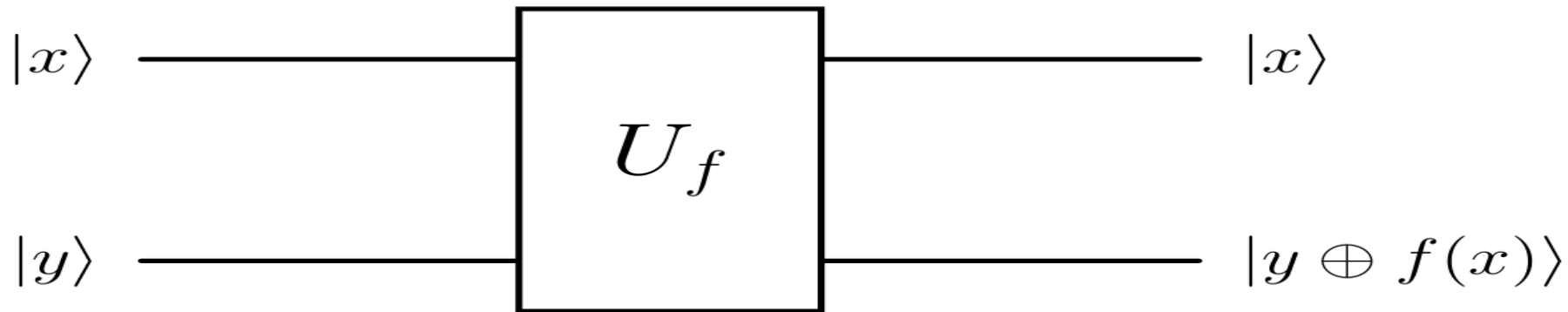
1 oracle query

The advantage comes from superposition and interference—not from reading every $f(x)$.

Oracle Representation

The oracle stores $f(x)$ reversibly

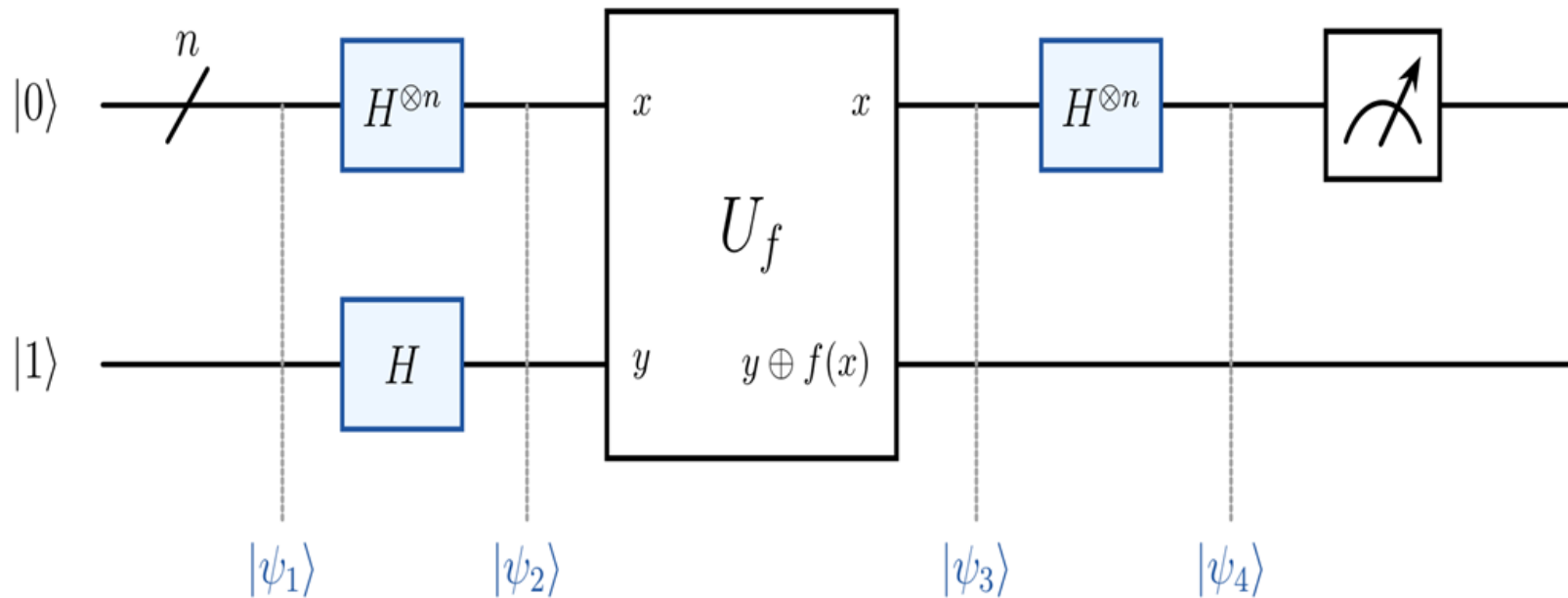
$$U_f : |x\rangle|y\rangle \longmapsto |x\rangle|y \oplus f(x)\rangle$$



x is unchanged; the ancilla changes from y to $y \oplus f(x)$. One passage through U_f is one oracle query.

Ideal Deutsch–Jozsa Circuit

The ideal circuit converts phase structure into a binary decision



Decision rule

Constant

$$P(0^n) = 1$$

Balanced

$$P(0^n) = 0$$

Circuit logic

Superposition $\rightarrow$ oracle phase encoding $\rightarrow$ interference $\rightarrow$ measurement

How One Query is Decoded

Interference creates the all-zero signature

1

Prepare

$|0\rangle^n|1\rangle$

2

Superpose

H creates all 2^n inputs

3

Encode

oracle adds phases $(-1)^{f(x)}$

4

Interfere

final H combines
amplitudes

Output

Constructive interference at $|0^n\rangle$ signals constant; cancellation at $|0^n\rangle$ signals balanced.

Rotation-Axis

Three coherent rotations model axis-specific gate miscalibration

$$R_x(\theta)$$

$$\exp(-i\theta X/2)$$

Bit-axis rotation

$$R_y(\theta)$$

$$\exp(-i\theta Y/2)$$

Mixed bit-phase rotation

$$R_z(\theta)$$

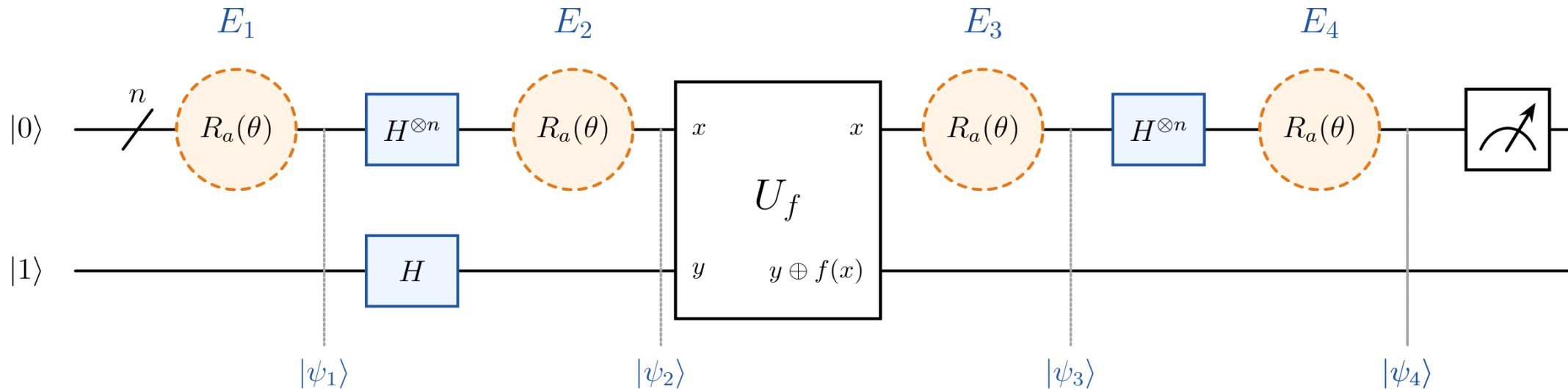
$$\exp(-i\theta Z/2)$$

Phase-axis rotation

Coherent error A fixed unitary over- or under-rotation; θ controls its magnitude and the axis controls its mechanism.

Localized Error Positions

Four locations isolate when the same rotation acts



- E_1 : Before the first Hadamard layer
- E_2 : After the first Hadamard layer and before U_f
- E_3 : After U_f and before the final Hadamard layer
- E_4 : After the final Hadamard layer and before measurement

Simulation Methodology

Error implementation

- Apply one localized rotation to one selected data qubit.
- Test $R_x(\theta)$, $R_y(\theta)$, and $R_z(\theta)$ separately.
- Repeat independently at E_1 , E_2 , E_3 , and E_4 .

Simulation parameters

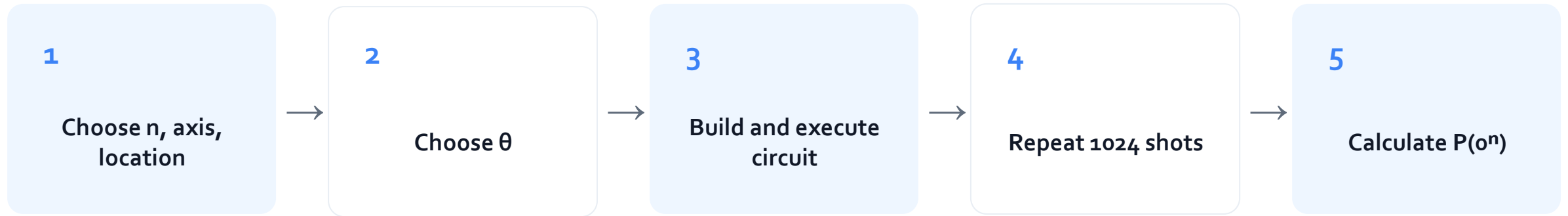
- Input qubits: $n \in \{1, 3, 5, 7, 9\}$; target $q = \lfloor n/2 \rfloor$.
- Angle: 0° – 180° in 20° increments.
- 1024 measurement shots per circuit execution.

A controlled sweep holds everything else fixed

Environment Python • NumPy • SciPy • Matplotlib • JupyterLab • Git

Simulation workflow

Every trial produces one comparable success probability



Performance metric

$$P(o^n) = N(o^n) / 1024$$

Fraction of shots that produce the all-zero data register

Results and Discussion

Sensitivity is an axis–location interaction

Nine plots test two constant functions and one balanced function across X, Y, and Z rotations.

X

E_1 and E_4

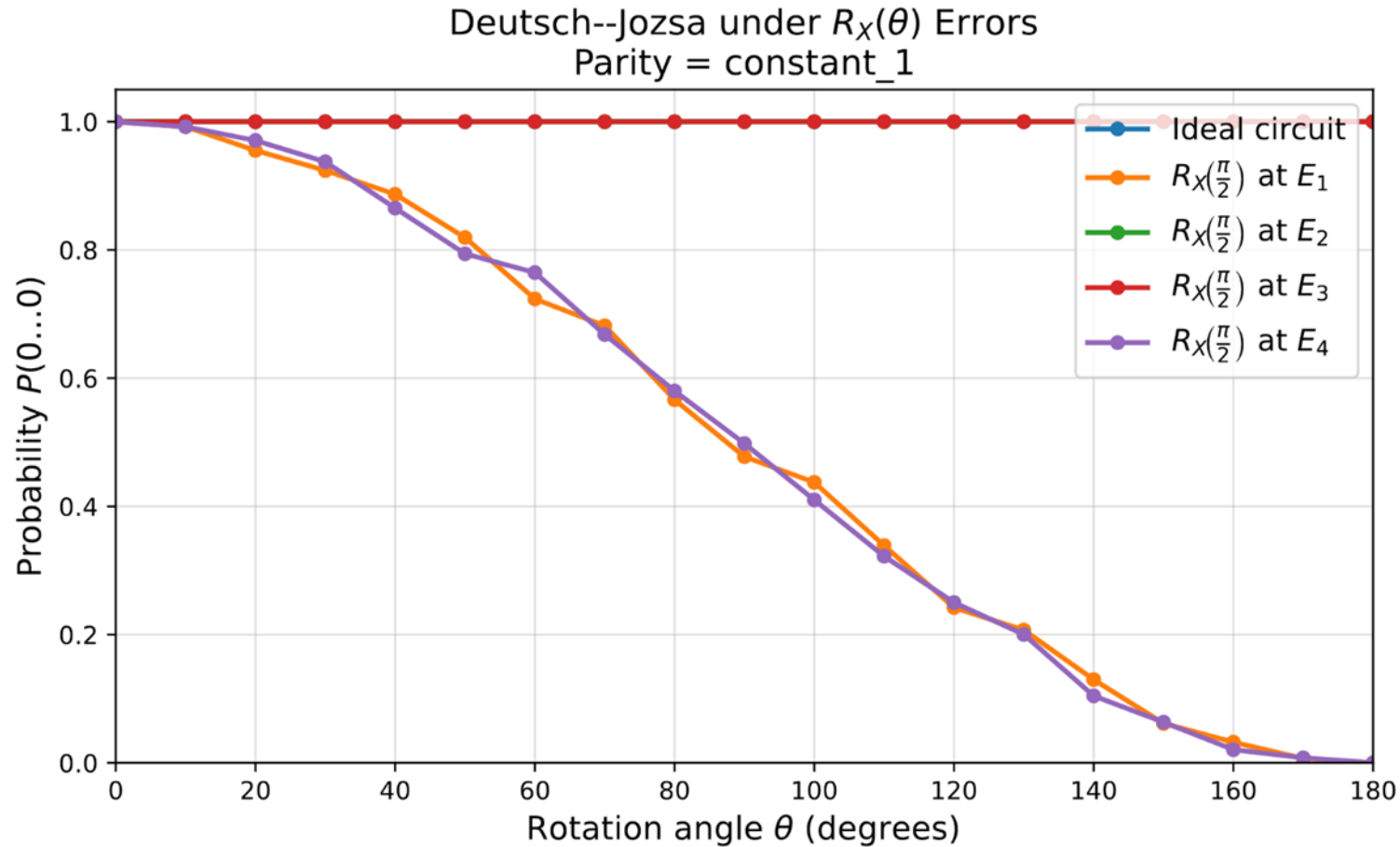
Y

All locations

Z

E_2 and E_3

Continued...



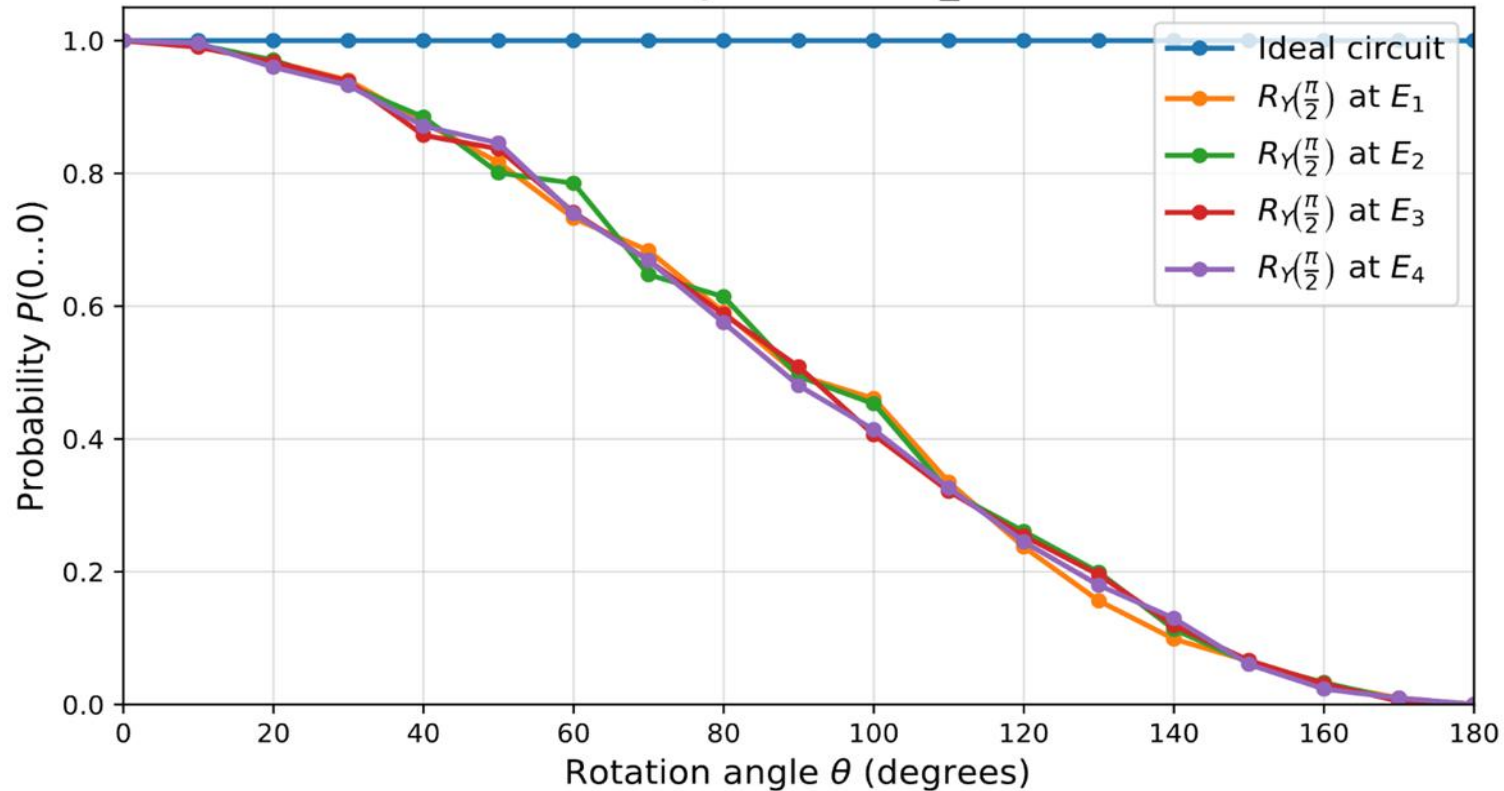
X-axis rotation

- E_1 and E_4 degrade together from $P(0^n) \approx 1$ toward 0.
- E_2 and E_3 remain close to the ideal curve.
- At intermediate stages, X-basis eigenstates turn the rotation mainly into phase.

**Most sensitive: state preparation
and pre-measurement**

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Deutsch--Jozsa under $R_Y(\theta)$ Errors
Parity = constant_1



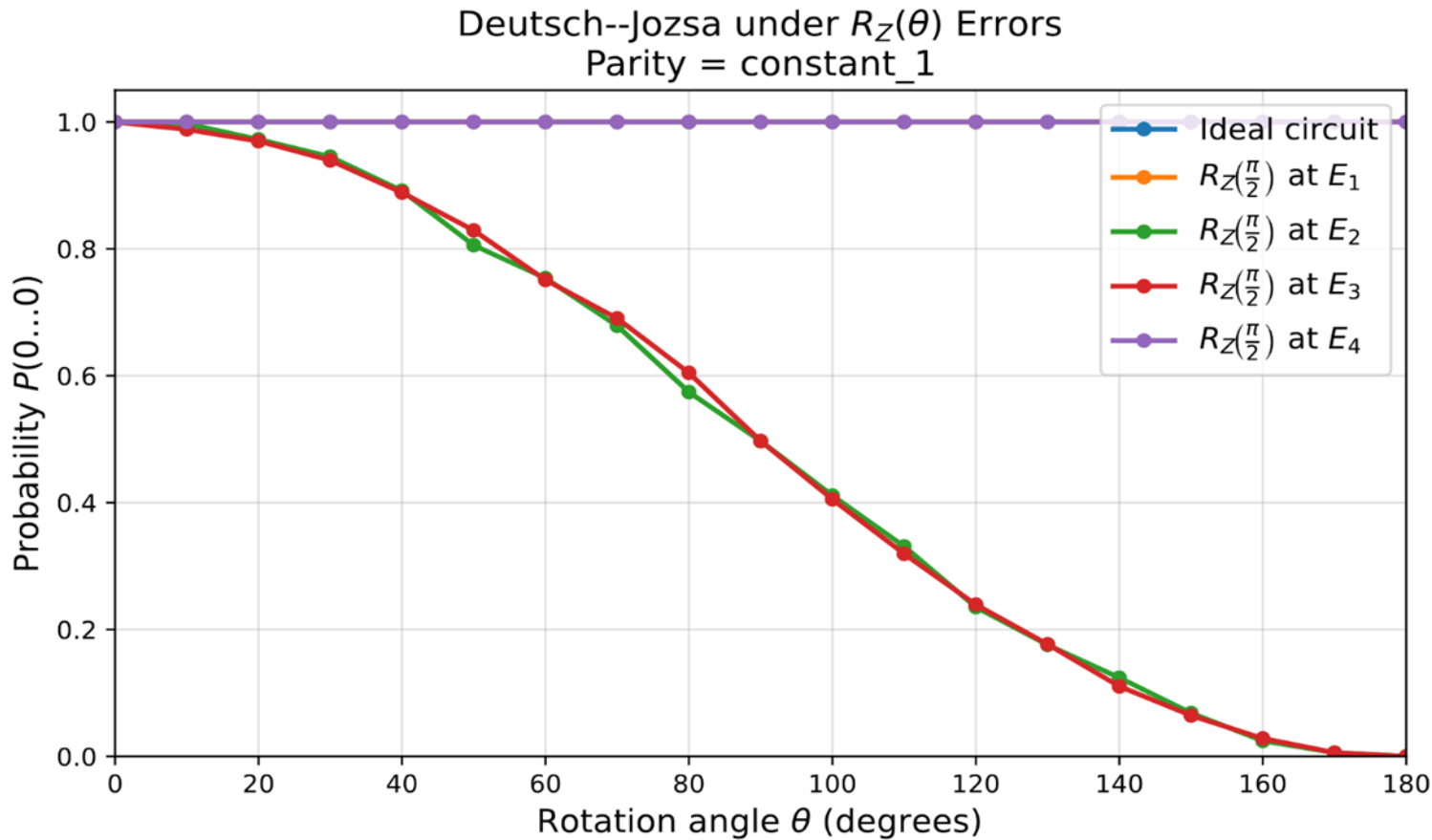
Each plotted rotation uses the angle sweep $\theta = 0^\circ \dots 180^\circ$

Y-axis rotation

- All four error-location curves nearly overlap.
- $P(0^n)$ decreases steadily as θ approaches 180° .
- Rotation angle dominates; no location is substantially protected.

Uniform location sensitivity

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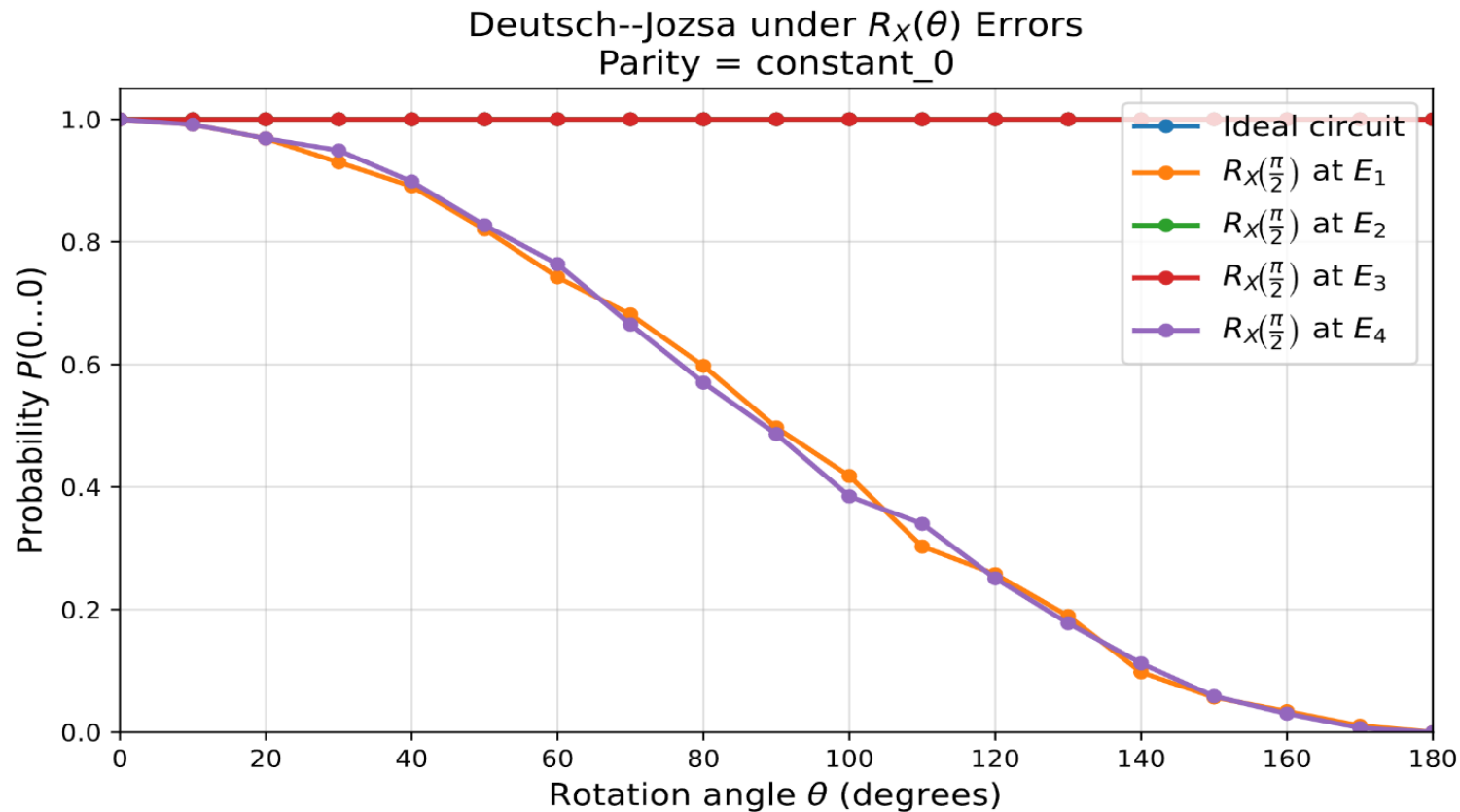
Each plotted rotation uses the angle sweep $\theta = 0^\circ \dots 180^\circ$

Z-axis rotation constant $f(x)=1$,
Z rotations target E_2 and E_3

- E_1 and E_4 remain essentially ideal.
- E_2 and E_3 lose success probability as θ increases.
- Relative phase changes inside the interference region alter the final outcome.

**Most sensitive: oracle input and
oracle output**

Continued...



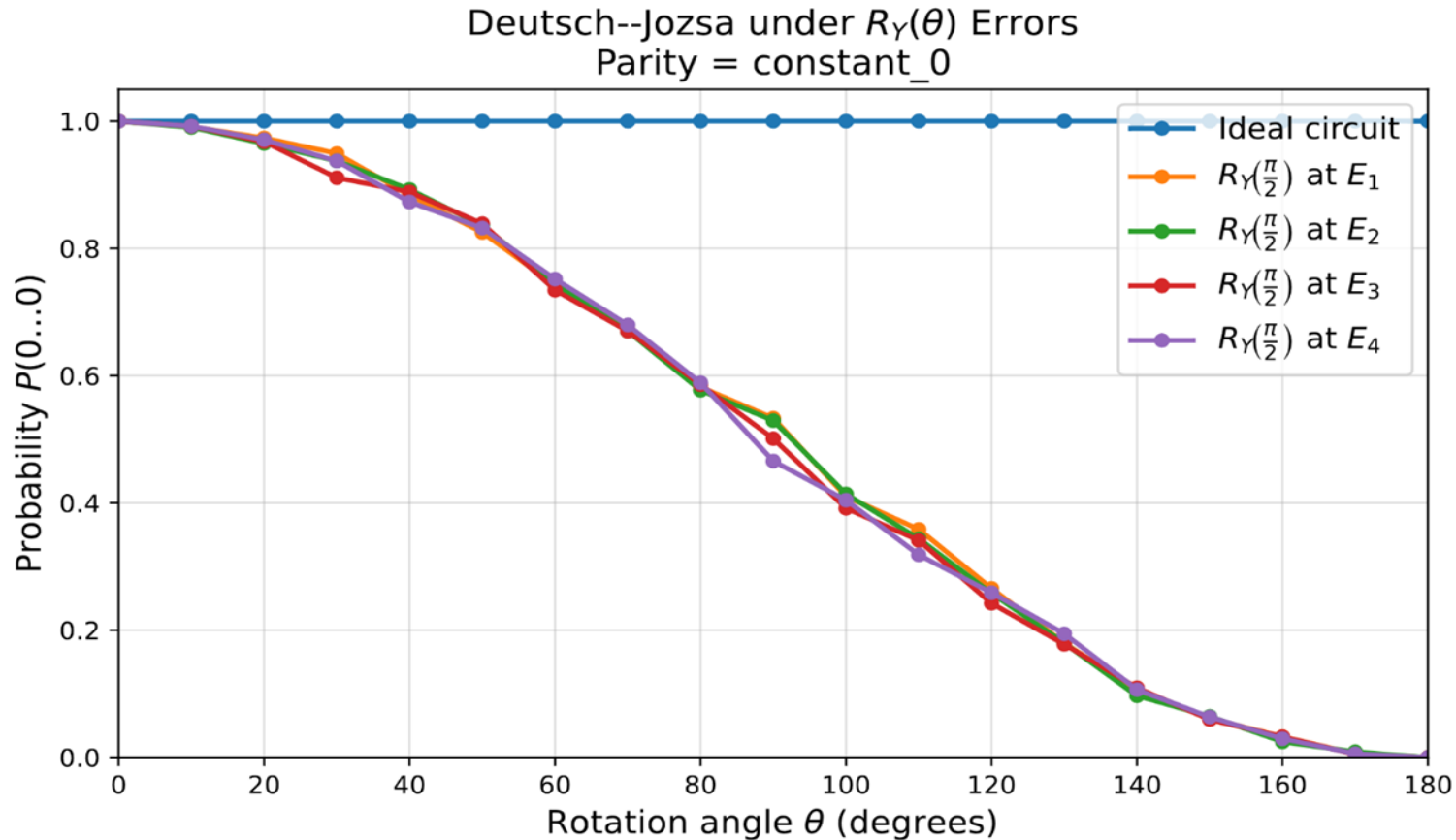
Each plotted rotation uses the angle sweep $\theta = 0^\circ \dots 180^\circ$

X-axis rotation · Constant $f(x)=0$

- E_1 and E_4 again show the dominant degradation.
- E_2 and E_3 remain near the ideal result.
- The location pattern is not specific to the constant output value.

Robust pattern across both constant functions

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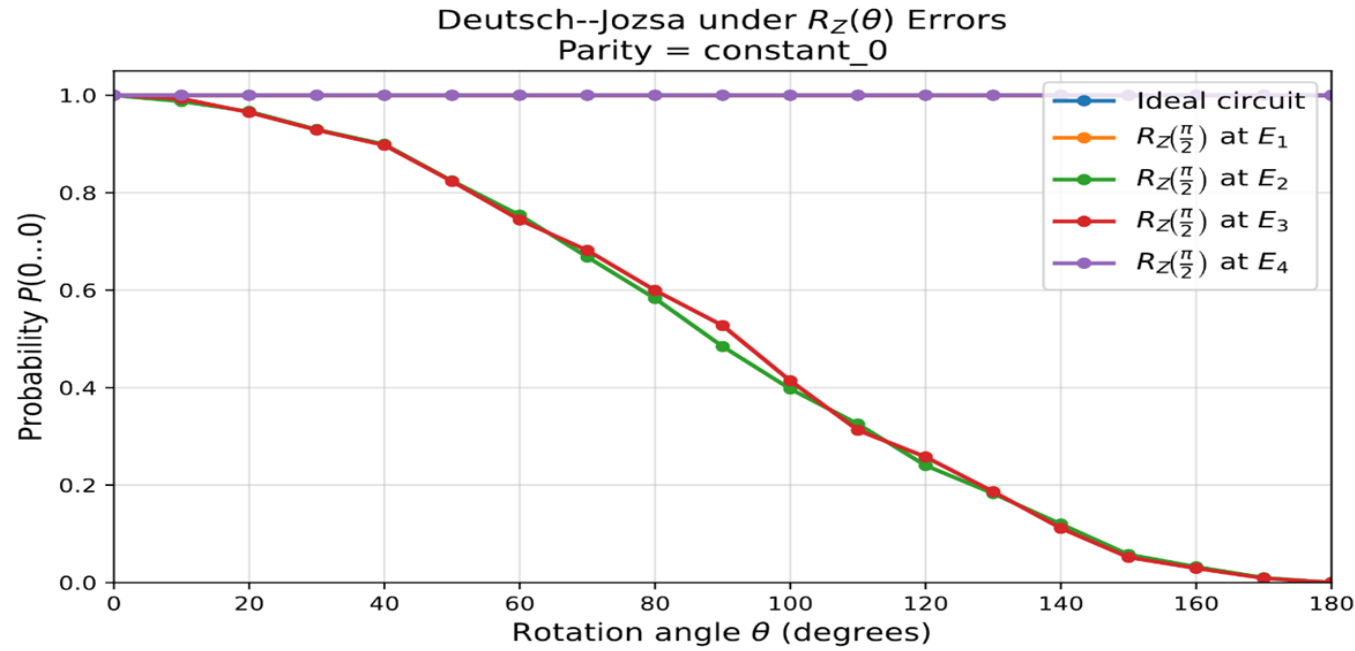
Each plotted rotation uses the angle sweep $\theta = 0^\circ \dots 180^\circ$

Y-axis rotation · Constant $f(x)=0$

- All locations decrease along nearly the same trajectory.
- Differences are consistent with finite-shot fluctuations.
- Angle remains more important than position for Y rotations.

Y response is location-independent

Continued...



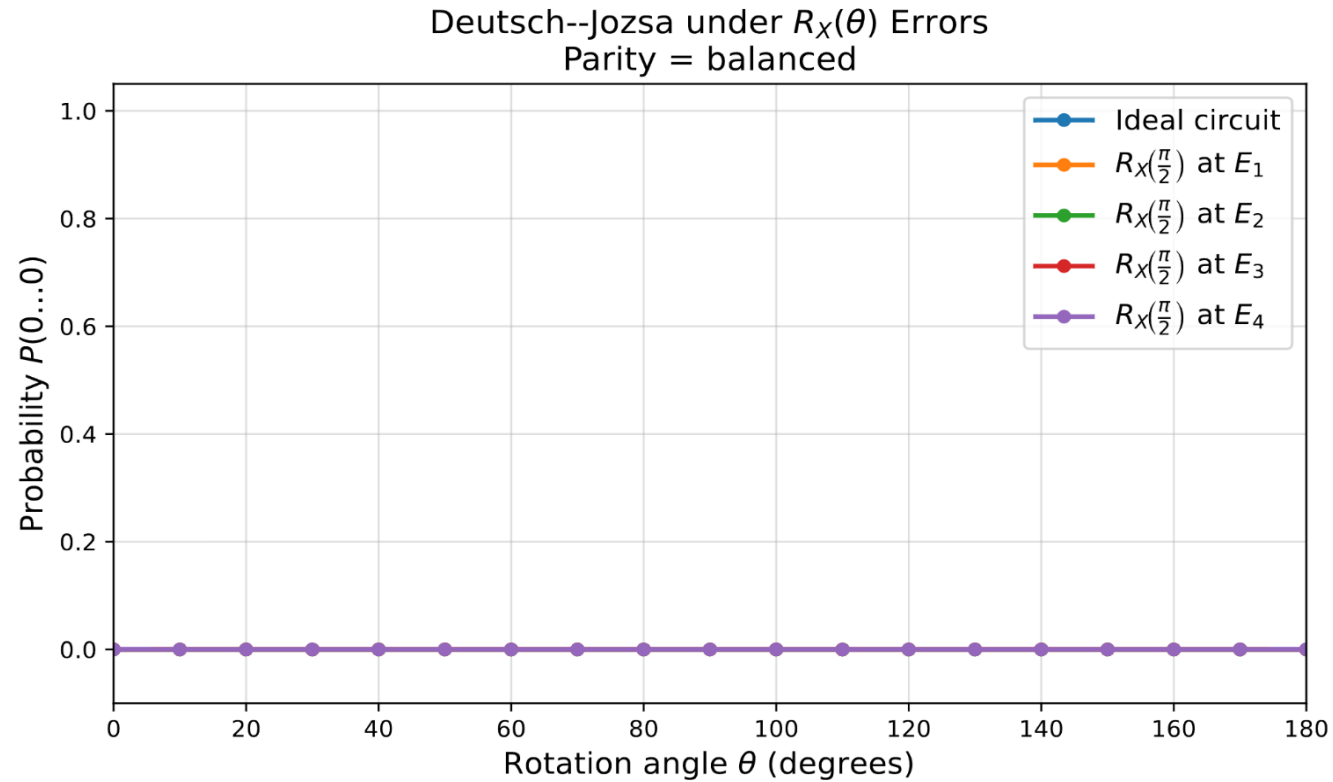
Z-axis rotation · constant o

- E_2 and E_3 are again the vulnerable positions.
- E_1 and E_4 stay near $P(0^n)=1$.
- The same phase-sensitive mechanism holds for both constant oracles.

Constant $f(x)=0$ reproduces the Z-axis interference sensitivity

Each plotted rotation uses the angle sweep $\theta = 0^\circ \dots 180^\circ$

Continued...

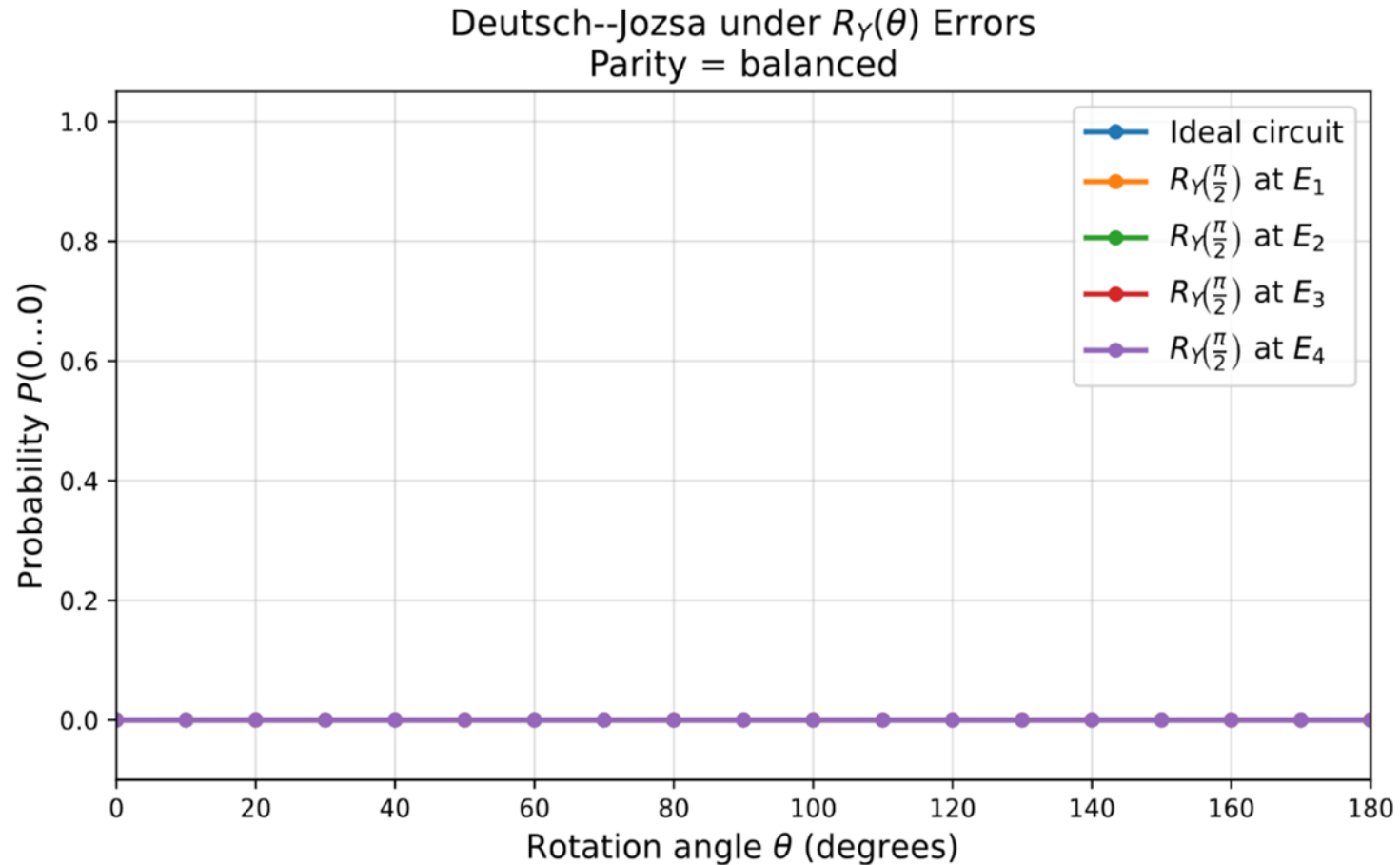


X-axis rotation · balanced

- $P(0^n)$ remains approximately zero for all locations and angles.
- All curves overlap with the ideal balanced result.
- This metric does not reveal changes among nonzero bit strings.

Each plotted rotation uses the angle sweep $\theta = 0^\circ \dots 180^\circ$

Continued...

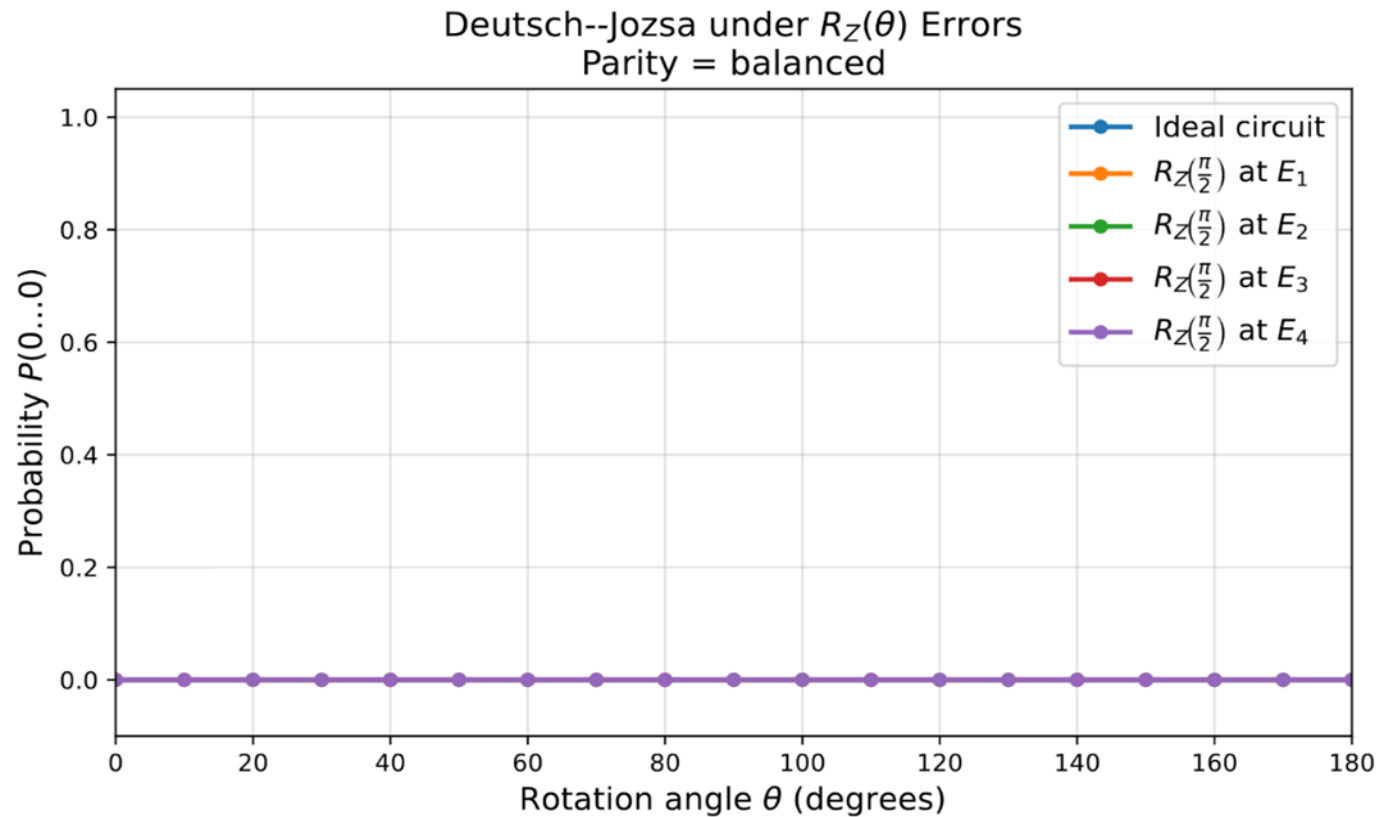


Each plotted rotation uses the angle sweep $\theta = 0^\circ \dots 180^\circ$

Y-axis rotation · balanced

- The all-zero probability stays at numerical zero.
- No axis–location dependence appears in $P(0^n)$.
- A full-distribution metric is needed to test whether other outputs changed.

Continued...



Each plotted rotation uses the angle sweep $\theta = 0^\circ \dots 180^\circ$

Z-axis rotation · balanced

- $P(0^n)$ remains approximately zero throughout the sweep.
- The decision rule still reports balanced.
- Stable classification does not imply that the complete output state is unchanged.

Cross-Result

Each axis has a distinct vulnerability map

Rotation

E_1

E_2

E_3

E_4

$R_x(\theta)$

HIGH

LOW

LOW

HIGH

$R_y(\theta)$

HIGH

HIGH

HIGH

HIGH

$R_z(\theta)$

LOW

HIGH

HIGH

LOW

Interpretation X errors attack circuit boundaries; Z errors attack the interference region; Y errors affect all stages.

Scope and Limitations

The conclusions are precise

What this study establishes

- Axis and circuit position jointly determine coherent-error sensitivity.
- The pattern repeats across both constant functions.
- Balanced classification is stable under the selected metric.

What it does not claim

- No depolarizing, damping, decoherence, or stochastic-noise model.
- One selected input qubit; no correlated or multi-qubit errors.
- $P(0^n)$ does not expose redistribution among nonzero outcomes.

Experimental scope

Simulation with 1024 shots; hardware drift and device-specific calibration are not modeled.

Conclusions

Error mitigation should be matched to the error mechanism

X

Protect preparation and pre-measurement stages

E_1 and E_4

Y

Treat the circuit as uniformly exposed

E_1 – E_4

Z

Protect the phase-sensitive interference region

E_2 and E_3

Thank You

Questions?